



Statistical Methods & Machine Learning 1

Class #7

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October 8, 2024

Outline

- A couple quick items
- Categorical Predictor Variables
 - Binary Predictor Variables
 - Linear Regression Intercepts
 - Predictors with More than 2 Categories
- Interactions of Predictor Variables

Exam 1

- Folks are still working on the exam this week due to different situations so please do not discuss it!
- We will start working on grading this week to try and get it turned around soon.

No class next week

- It will be fall break for UM next Monday/Tuesday so there will be **no class for anyone on October 15.**
- This means you will have an extra week to complete HW5.
 - Due by 9:30am on October 22
- Homework groups will change starting with HW 6



Interpreting regression coefficients

Linear regression with continuous predictor(s)

- Without any predictor

$$y_i = \beta_0 + \epsilon_i$$

- One continuous predictor

$$y_i = \beta_0 + \beta_1 x_1 + \epsilon_i$$

- Multiple continuous predictors

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon_i$$



Categorical Predictor Variables

Predictor Variable Types

- Quantitative
 - Continuous (e.g., Income in \$)
 - Ordinal (e.g., Strongly Agree – Agree – Neither – Disagree – Strongly Disagree)
 - Likert Scale (pronounced Lick-ert)
- Qualitative
 - Two categories (e.g., Yes, No); Binary; Dichotomous; Bernoulli
 - More than two categories (e.g., Race: Black, White, American Indian, Asian, Other); Categorical; Multinomial

Let's look at some data: Wage in ISLR - 1

A data frame with 3000 observations on the following 11 variables.

`year` Year that wage information was recorded

`age` Age of worker

`maritl` A factor with levels 1. Never Married 2. Married 3. Widowed 4. Divorced and 5. Separated indicating marital status

`race` A factor with levels 1. White 2. Black 3. Asian and 4. Other indicating race

`education` A factor with levels 1. < HS Grad 2. HS Grad 3. Some College 4. College Grad and 5. Advanced Degree indicating education level

`region` Region of the country (mid-atlantic only)

`jobclass` A factor with levels 1. Industrial and 2. Information indicating type of job

`health` A factor with levels 1. <=Good and 2. >=Very Good indicating health level of worker

`health_ins` A factor with levels 1. Yes and 2. No indicating whether worker has health insurance

`logwage` Log of workers wage

`wage` Workers raw wage

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race A factor with levels 1. White 2. Black 3. Asian and 4. Other indicating race

education A factor with levels 1. < HS Grad 2. HS Grad 3. Some College 4. College Grad and 5. Advanced Degree indicating education level

region Region of the country (mid-atlantic only)

jobclass A factor with levels 1. Industrial and 2. Information indicating type of job

health A factor with levels 1. <=Good and 2. >=Very Good indicating health level of worker

health_ins A factor with levels 1. Yes and 2. No indicating whether worker has health insurance

logwage Log of workers wage

wage Workers raw wage

Let's look at some data: Wage in ISLR - 1

A data frame with 3000 observations on the following 11 variables.

year Year that wage information was recorded

age Age of worker

maritl A factor with levels 1. Married, 2. Single, 3. Divorced and 4. Separated in

race A factor with levels 1. White, 2. Black, 3. Asian and 4. Other

education A factor with levels 1. Less than High School, 2. High School, 3. Some College, 4. College Grad and 5. Advanced

region Region of the country (mid-atlantic only)

jobclass A factor with levels 1. Industrial and 2. Information indicating type of job

health A factor with levels 1. <=Good and 2. >=Very Good indicating health level of worker

health_ins A factor with levels 1. Yes and 2. No indicating whether worker has health insurance

logwage Log of workers wage

wage Workers raw wage

Allows us to explore questions like:
"how does the expected mean
wage change with a one-year
change in age."

Let's look at some data: Wage in ISLR - 1

A data frame with 3000 observations on the following 11 variables.

year Year that wage information was recorded

age Age of worker

maritl A factor with
5. Separated in

race A factor with levels

education A factor with
and 5. Advanced

region Region of the country (mid-atlantic only)

jobclass A factor with levels 1. Industrial and 2. Information indicating type of job

health A factor with

health_ins A factor

logwage Log of worker's wage

wage Workers raw wage

4. Divorced and

ing race

4. College Grad

What about if we wanted to explore the association between wage and a non-continuous predictor like "jobclass"?

How does that impact the way we interpret our model?

What if we wanted to look at a categorical predictor with >2 categories like "education"?



Binary Predictor Variables

Binary Predictor Variables

$y_1, \dots, y_n$ from the combined sample of 2 levels (e.g., wage from industrial vs. information job classes)

Two dummy variables (d_{1i}, d_{2i}) defined as:

$$d_{1i} = \begin{cases} 1 & \text{If } i^{th} \text{ unit is in level 1 (e.g., male); treatment} \\ 0 & \text{otherwise; reference} \end{cases}$$

$$d_{2i} = \begin{cases} 1 & \text{If } i^{th} \text{ unit is in level 2 (e.g., female); treatment} \\ 0 & \text{otherwise; reference} \end{cases}$$

Hence, $d_{1i} + d_{2i} = 1$

Binary Predictor Variables

Let's look at an example using the `Wage` dataset from the `ISLR2` package. We'll fit a simple linear regression model with `wage` as a function of `jobclass`.

We'll create our dummy variables manually in this case. Specifically:

- `indus`: 1 = industrial; 0 = non-industrial
- `info`: 1 = information; 0 = non-information

```
lm(wage ~ indus + info, Wage)
```

Binary Predictor Variables

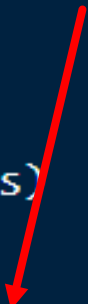
```
Call:
lm(formula = wage ~ indus + info, data = wage)

Residuals:
    Min       1Q   Median       3Q      Max
-100.507  -25.362   -6.117   15.697  197.750

Coefficients: (1 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   120.593      1.070   112.69  <2e-16 ***
indus         -17.272      1.492   -11.58  <2e-16 ***
info           NA          NA      NA      NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.83 on 2998 degrees of freedom
Multiple R-squared:  0.04281,    Adjusted R-squared:  0.04249
F-statistic: 134.1 on 1 and 2998 DF,  p-value: < 2.2e-16
```

What happened here?



Binary Predictor Variables

```
Call:
lm(formula = wage ~ indus + info, data = wage)

Residuals:
    Min       1Q   Median       3Q      Max
-100.507  -25.362   -6.117   15.697  197.750

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(Intercept)  120.593      1.070   112.69  <2e-16 ***
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info          NA          NA      NA      NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

What happened here?

```
Residual standard error: 40.83 on 2998 degrees of freedom
Multiple R-squared:  0.04281,    Adjusted R-squared:  0.04176
F-statistic: 134.1 on 1 and 2998 DF,  p-value: < 2.2e-16
```

The model.matrix() function can give us a clue.

Binary Predictor Variables

Y (Income)	X_1 (Education)	X_2 (Seniority)
99.917	21.586	113.103
92.579	18.276	119.310
34.679	12.069	100.690
$\vdots$	$\vdots$	$\vdots$
74.611	17.034	106.897

`model.matrix()`
gives us the
observed values
used in the model
which multiplied by
their respective
betas and added
together give us the
fitted values

$$\begin{pmatrix} 99.917 \\ 92.579 \\ 34.679 \\ \vdots \\ 74.611 \end{pmatrix} = \beta_0 + \beta_1 \times \begin{pmatrix} 21.586 \\ 18.276 \\ 12.069 \\ \vdots \\ 17.034 \end{pmatrix} + \beta_2 \times \begin{pmatrix} 113.103 \\ 119.310 \\ 100.690 \\ \vdots \\ 106.897 \end{pmatrix} + \begin{pmatrix} \epsilon_2 \\ \epsilon_3 \\ \vdots \\ \epsilon_N \end{pmatrix}$$

Gives a vector of 1's for the intercept as $\beta_0 * 1 = \beta_0$

Binary Predictor Variables

```
> head(model.matrix(mod_j1))
```

	(Intercept)	indus	info
231655	1	1	0
86582	1	0	1
161300	1	1	0
155159	1	0	1
11443	1	0	1
376662	1	0	1

Both of our predictors are dummy/indicator variables derived from the `jobclass` variable.

Does anything stand out?

Binary Predictor Variables

```
> head(model.matrix(mod_j1))  
      (Intercept)  induc  info  
231655           1      1     0  
86582            1      0     1  
161300           1      1     0  
155159           1      0     1  
11443            1      0     1  
376662           1      0     1
```

Both of our predictors are dummy/indicator variables derived from the `jobclass` variable.

Does anything stand out?

They're simply the inverse of each other. No additional information is provided by one that's not already provided by the other. We're trying to use 3 parameters to model only 2 groups.

Binary Predictor Variables

```
> head(model.matrix(mod_j1))
      (Intercept)  indus  info
231655           1      1     0
86582            1      0     1
161300           1      1     0
155155           1      0     1
11443            1      0     1
376662           1      0     1
```

Both of our predictors are dummy/indicator variables derived from the `jobclass` variable.

We would say that this matrix is “not of full rank”

Does anything stand out?

They’re simply the inverse of each other. No additional information is provided by one that’s not already provided by the other. We’re trying to use 3 parameters to model only 2 groups.

Dummy Variables in Regression - 3

Non-estimable model: $y_i = \beta_0 + \beta_1 d_{1i} + \beta_2 d_{2i} + \epsilon_i$

Two ways to make the model estimable:

1. Drop the intercept
2. Drop one of the dummy variables
 - Also viewed through linear transformation

$$\begin{aligned} y_i &= \mu_1 d_{1i} + \mu_2 d_{2i} + \epsilon_i \\ &= \mu_1 (1 - d_{2i}) + \mu_2 d_{2i} + \epsilon_i \\ &= \mu_1 + (\mu_2 - \mu_1) d_{2i} + \epsilon_i \\ &= \gamma_0 + \gamma_2 d_{2i} + \epsilon_i \end{aligned}$$

- Inferences about γ_2 allows hypothesis test as
 $H_0: \gamma_2 = 0 (\rightarrow \mu_1 = \mu_2)$ vs. $H_A: \gamma_2 \neq 0 (\rightarrow \mu_1 \neq \mu_2)$

What does γ_2 mean?
What about γ_0 ?
How about $\gamma_0 + \gamma_2$?

Dummy Variables in Regression - 4

Non-estimable model: $y_i = \beta_0 + \beta_1 d_{1i} + \beta_2 d_{2i} + \epsilon_i$

Two ways to make the model estimable:

1. Drop the intercept
2. Drop one of the dummy variables
 - Also viewed through linear transformation

$$\begin{aligned} y_i &= \mu_1 d_{1i} + \mu_2 d_{2i} + \epsilon_i \\ &= \mu_1 (1 - d_{2i}) + \mu_2 d_{2i} + \epsilon_i \\ &= \mu_1 + (\mu_2 - \mu_1) d_{2i} + \epsilon_i \\ &= \gamma_0 + \gamma_2 d_{2i} + \epsilon_i \end{aligned}$$

- Inferences about γ_2 allows hypothesis test as
 $H_0: \gamma_2 = 0 (\rightarrow \mu_1 = \mu_2)$ vs. $H_A: \gamma_2 \neq 0 (\rightarrow \mu_1 \neq \mu_2)$

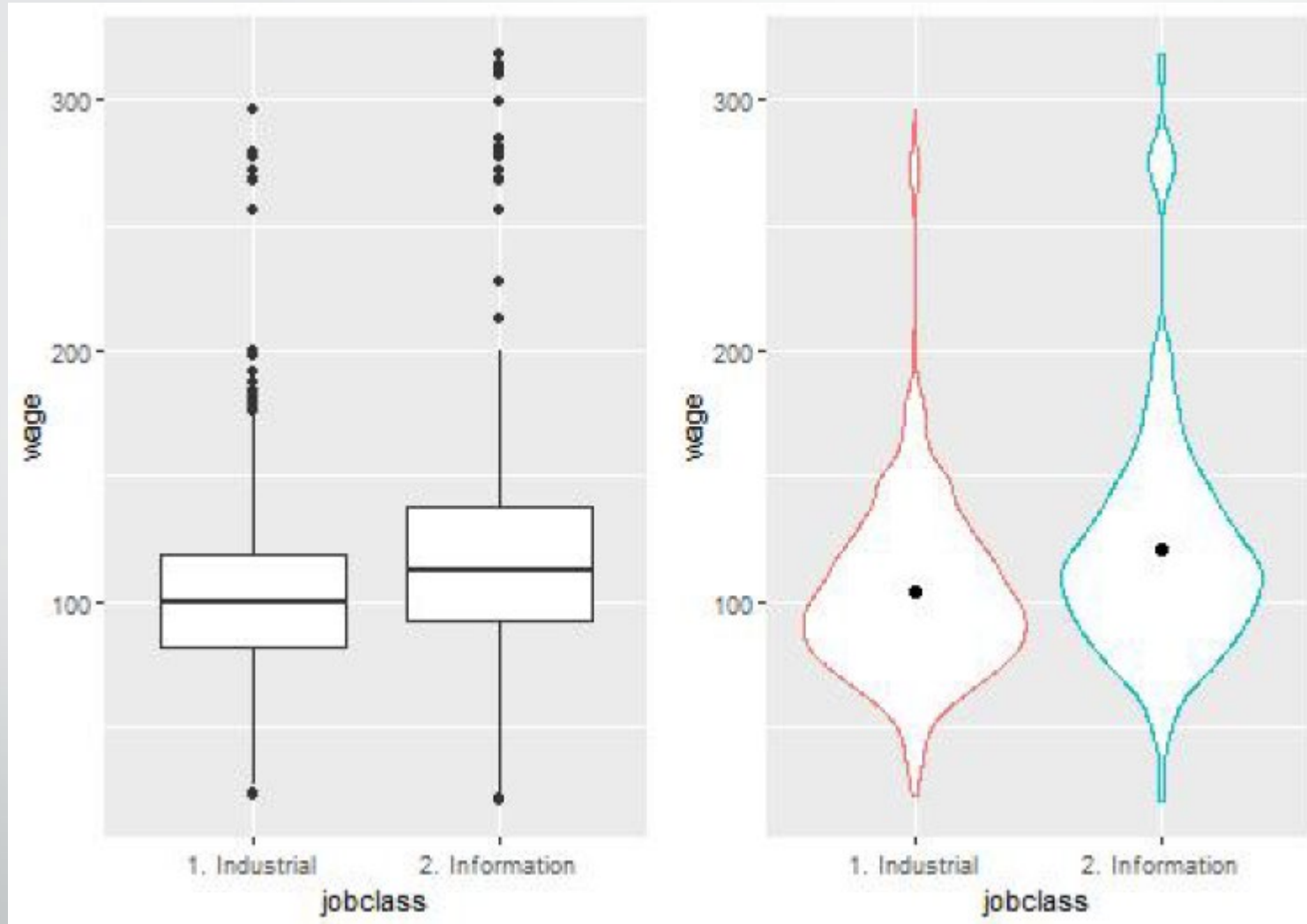
R will do this automatically when you fit a model with a “factor” predictor. This becomes your reference group.

What does γ_2 mean?
What about γ_0 ?
How about $\gamma_0 + \gamma_2$?

Let's look again at Wage in ISLR2 - 1

	year	age	maritl	race	education	region	jobclass	health	health_ins	logwage	wage
231655	2006	18	1. Never Married	1. White	1. < HS Grad	2. Middle Atlantic	1. Industrial	1. <=Good	2. No	4.318063	75.04315
86582	2004	24	1. Never Married	1. White	4. College Grad	2. Middle Atlantic	2. Information	2. >=Very Good	2. No	4.255273	70.47602
161300	2003	45	2. Married	1. White	3. Some College	2. Middle Atlantic	1. Industrial	1. <=Good	1. Yes	4.875061	130.98218
155159	2003	43	2. Married	3. Asian	4. College Grad	2. Middle Atlantic	2. Information	2. >=Very Good	1. Yes	5.041393	154.68529
11443	2005	50	4. Divorced	1. White	2. HS Grad	2. Middle Atlantic	2. Information	1. <=Good	1. Yes	4.318063	75.04315
376662	2008	54	2. Married	1. White	4. College Grad	2. Middle Atlantic	2. Information	2. >=Very Good	1. Yes	4.845098	127.11574
450601	2009	44	2. Married	4. Other	3. Some College	2. Middle Atlantic	1. Industrial	2. >=Very Good	1. Yes	5.133021	169.52854
377954	2008	30	1. Never Married	3. Asian	3. Some College	2. Middle Atlantic	2. Information	1. <=Good	1. Yes	4.716003	111.72085
228963	2006	41	1. Never Married	2. Black	3. Some College	2. Middle Atlantic	2. Information	2. >=Very Good	1. Yes	4.778151	118.88436
81404	2004	52	2. Married	1. White	2. HS Grad	2. Middle Atlantic	2. Information	2. >=Very Good	1. Yes	4.857332	128.68049
302778	2007	45	4. Divorced	1. White	3. Some College	2. Middle Atlantic	2. Information	1. <=Good	1. Yes	4.763428	117.14682
305706	2007	34	2. Married	1. White	2. HS Grad	2. Middle Atlantic	1. Industrial	2. >=Very Good	2. No	4.397940	81.28325
8690	2005	35	1. Never Married	1. White	2. HS Grad	2. Middle Atlantic	2. Information	2. >=Very Good	1. Yes	4.494155	89.49248
153561	2003	39	2. Married	1. White	4. College Grad	2. Middle Atlantic	1. Industrial	2. >=Very Good	1. Yes	4.903090	134.70538
449654	2009	54	2. Married	1. White	2. HS Grad	2. Middle Atlantic	2. Information	2. >=Very Good	1. Yes	4.903090	134.70538
447660	2009	51	2. Married	1. White	3. Some College	2. Middle Atlantic	1. Industrial	2. >=Very Good	1. Yes	4.505150	90.48191
160191	2003	37	1. Never Married	3. Asian	4. College Grad	2. Middle Atlantic	1. Industrial	2. >=Very Good	2. No	4.414973	82.67964
230312	2006	50	2. Married	1. White	5. Advanced Degree	2. Middle Atlantic	2. Information	2. >=Very Good	2. No	5.360552	212.84235
301585	2007	56	2. Married	1. White	4. College Grad	2. Middle Atlantic	1. Industrial	1. <=Good	1. Yes	4.861026	129.15669
153682	2003	37	1. Never Married	1. White	3. Some College	2. Middle Atlantic	1. Industrial	2. >=Very Good	1. Yes	4.591065	98.59934
158226	2003	38	2. Married	3. Asian	4. College Grad	2. Middle Atlantic	2. Information	2. >=Very Good	2. No	5.301030	200.54326
11141	2005	40	4. Divorced	1. White	2. HS Grad	2. Middle Atlantic	2. Information	2. >=Very Good	2. No	3.920123	50.40666

Let's look again at Wage in ISLR2 - 2



Model 1 – Wage and jobclass

- Can write this model as:

```
mod_job <- lm(wage ~ jobclass, Wage)
```

```
> summary(mod_job)

Call:
lm(formula = wage ~ jobclass, data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-100.507  -25.362   -6.117   15.697  197.750

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    103.321     1.039   99.43  <2e-16 ***
jobclass2. Information  17.272     1.492   11.58  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.83 on 2998 degrees of freedom
Multiple R-squared:  0.04281, Adjusted R-squared:  0.04249
F-statistic: 134.1 on 1 and 2998 DF, p-value: < 2.2e-16
```

What do you notice?

Model 1 – Wage and jobclass

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```
mod_job <- lm(wage ~ jobclass, Wage)
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> summary(mod_job)

Call:
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Residuals:
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```

How would you interpret the coefficients?

Model 1 – Wage and jobclass

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```
mod_job <- lm(wage ~ jobclass, Wage)
```

```
> summary(mod_job)
```

Call:

```
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Residuals:

Min	1Q	Median	3Q	Max
-100.507	-25.362	-6.117	15.697	197.750

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	103.321	1.039	99.43	<2e-16 ***
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Mean expected raw wage for workers
whose jobclass = 1. Industrial

Model 1 – Wage and jobclass

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```
mod_job <- lm(wage ~ jobclass, Wage)
```

```
> summary(mod_job)

Call:
lm(formula = wage ~ jobclass, data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-100.507  -25.362   -6.117   15.697  197.750

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    103.321     1.039   99.43  <2e-16 ***
jobclass2. Information 17.272     1.492   11.58  <2e-16 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.83 on 2998 degrees of freedom
Multiple R-squared:  0.04281,    Adjusted R-squared:  0.04249
F-statistic: 134.1 on 1 and 2998 DF,  p-value: < 2.2e-16
```

Difference in mean expected raw wage for workers with Information jobclass vs. industrial jobclass.

Model 1 – Wage and jobclass

- Can write this model as:

```
mod_job <- lm(wage ~ jobclass, Wage)
```

```
> summary(mod_job)
```

Call:

```
lm(formula = wage ~ jobclass, data = Wage)
```

Residuals:

Min	1Q	Median	3Q	Max
-100.507	-25.362	-6.117	15.697	197.750

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	103.321	1.039	99.43	<2e-16 ***
jobclass2. Information	17.272	1.492	11.58	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.83 on 2998 degrees of freedom

Multiple R-squared: 0.04281, Adjusted R-squared: 0.04249

F-statistic: 134.1 on 1 and 2998 DF, p-value: < 2.2e-16

How would we get the mean expected raw wage for information workers?

Model 1 – Wage and jobclass

- Can write this model as:

```
mod_job <- lm(wage ~ jobclass, Wage)
```

```
> summary(mod_job)
```

Call:

```
lm(formula = wage ~ jobclass, data = Wage)
```

Residuals:

Min	1Q	Median	3Q	Max
-100.507	-25.362	-6.117	15.697	197.75

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	103.321	1.039	99.43	<2e-16 ***
jobclass2. Information	17.272	1.492	11.58	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.83 on 2998 degrees of freedom

Multiple R-squared: 0.04281, Adjusted R-squared: 0.04249

F-statistic: 134.1 on 1 and 2998 DF, p-value: < 2.2e-16

How would we get the mean expected raw wage for information workers?

$$\beta_0 + \beta_1 = 103.32 + 17.27 = 120.59$$

Model 1 – Wage and jobclass

- Can write this model as:

```
mod_job <- lm(wage ~ jobclass, Wage)
```

```
> summary(mod_job)

Call:
lm(formula = wage ~ jobclass, data = Wage)

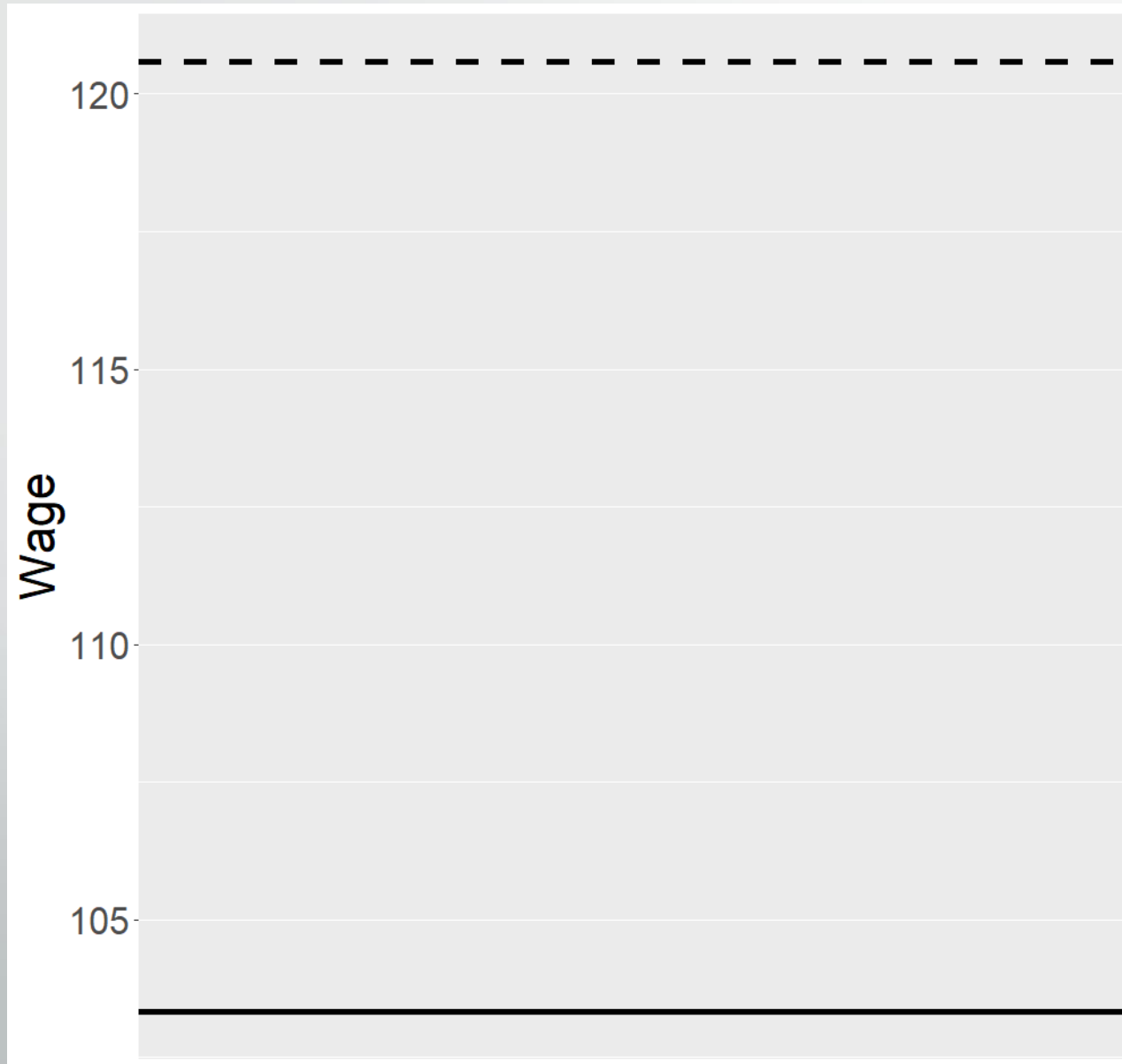
Residuals:
    Min       1Q   Median       3Q      Max
-100.507  -25.362   -6.117   15.697  197.750

Coefficients:
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Multiple R-squared:  0.04281, Adjusted R-squared:  0.04249
F-statistic: 134.1 on 1 and 2998 DF, p-value: < 2.2e-16
```

What does this look like graphically?

Model 1 – Wage and jobclass



$\beta_0 + \beta_1$: Mean Wage for those in information jobs

β_0 : Mean Wage for those in industrial jobs



Model with categorical and continuous
predictors

Model 2 – Wage, jobclass, and age

- Can write this model as:

```
mod_jobage <- lm(wage ~ jobclass + age, Wage)
```

```
> summary(mod_jobage)

Call:
lm(formula = wage ~ jobclass + age, data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-107.234  -24.751   -6.311   16.308   197.278

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    76.6298     2.8320  27.06  <2e-16 ***
jobclass2. Information 15.9214     1.4732  10.81  <2e-16 ***
age              0.6447     0.0638  10.11  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.16 on 2997 degrees of freedom
Multiple R-squared:  0.07435,    Adjusted R-squared:  0.07373
F-statistic: 120.4 on 2 and 2997 DF,  p-value: < 2.2e-16
```

Model 2 – Wage, jobclass, and age

- Can write this model as:

```
mod_jobage <- lm(wage ~ jobclass + age, Wage)
```

```
> summary(mod_jobage)

Call:
lm(formula = wage ~ jobclass + age, data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-107.234  -24.751   -6.311   16.308   197.278

Coefficients:
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Residual standard error: 40.16 on 2997 degrees of freedom
Multiple R-squared:  0.07435,    Adjusted R-squared:  0.07373
F-statistic: 120.4 on 2 and 2997 DF,  p-value: < 2.2e-16
```

How would you interpret the coefficients?

Model 2 – Wage, jobclass, and age

- Can write this model as:

```
mod_jobage <- lm(wage ~ jobclass + age, Wage)
```

```
> summary(mod_jobage)
```

Call:

```
lm(formula = wage ~ jobclass + age, data = Wa
```

Residuals:

Min	1Q	Median	3Q	Max
-107.234	-24.751	-6.311	16.308	197.278

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	76.6298	2.8320	27.06	<2e-16 ***
jobclass2. Information	15.9214	1.4732	10.81	<2e-16 ***
age	0.6447	0.0638	10.11	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.16 on 2997 degrees of freedom

Multiple R-squared: 0.07435, Adjusted R-squared: 0.07373

F-statistic: 120.4 on 2 and 2997 DF, p-value: < 2.2e-16

Mean expected raw wage for workers aged 0 and industrial jobclass.

Model 2 – Wage, jobclass, and age

- Can write this model as:

```
mod_jobage <- lm(wage ~ jobclass + age, Wage)
```

```
> summary(mod_jobage)

Call:
lm(formula = wage ~ jobclass + age, data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-107.234  -24.751   -6.311   16.308  197.278

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    76.6298     2.8320   27.06  <2e-16 ***
jobclass2. Information 15.9214     1.4732   10.81  <2e-16 ***
age             0.6447     0.0638   10.11  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.16 on 2997 degrees of freedom
Multiple R-squared:  0.07435,    Adjusted R-squared:  0.07373
F-statistic: 120.4 on 2 and 2997 DF,  p-value: < 2.2e-16
```

Difference in mean expected raw wage for Information workers vs industrial workers while holding age constant.

Model 2 – Wage, jobclass, and age

- Can write this model as:

```
mod_jobage <- lm(wage ~ jobclass + age, Wage)
```

```
> summary(mod_jobage)

Call:
lm(formula = wage ~ jobclass + age, data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-107.234  -24.751   -6.311   16.308   197.278

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    76.6298     2.8320  27.06  <2e-16 ***
jobclass2. Information 15.9214     1.4732  10.81  <2e-16 ***
age              0.6447     0.0638  10.11  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.16 on 2997 degrees of freedom
Multiple R-squared:  0.07435,    Adjusted R-squared:  0.07373
F-statistic: 120.4 on 2 and 2997 DF,  p-value: < 2.2e-16
```

One unit increase in age is associated with 0.65 increase in expected mean wage while keeping jobclass constant.

Model 2 – Wage, jobclass, and age

- Can write this model as:

```
mod_jobage <- lm(wage ~ jobclass + age, Wage)
```

```
> summary(mod_jobage)

Call:
lm(formula = wage ~ jobclass + age, data = Wage)

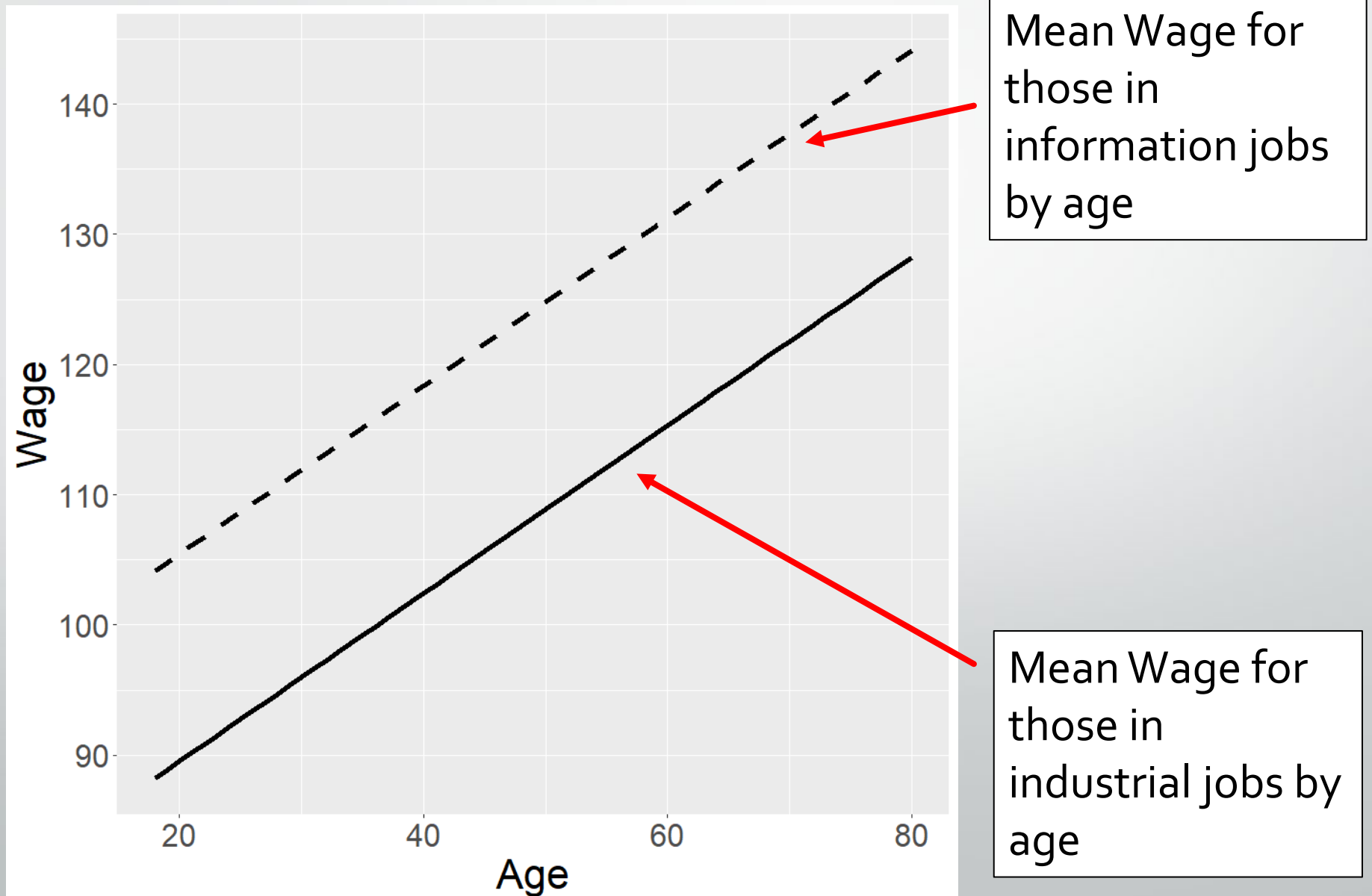
Residuals:
    Min       1Q   Median       3Q      Max
-107.234  -24.751   -6.311   16.308   197.278

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    76.6298     2.8320   27.06  <2e-16 ***
jobclass2. Information 15.9214     1.4732   10.81  <2e-16 ***
age              0.6447     0.0638   10.11  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.16 on 2997 degrees of freedom
Multiple R-squared:  0.07435,    Adjusted R-squared:  0.07373
F-statistic: 120.4 on 2 and 2997 DF,  p-value: < 2.2e-16
```

What does this
look like
graphically?

Model 2 – Wage, jobclass, and age





Multiple categorical predictors

Model 2 – Wage, jobclass, and health_ins

- Can write this model as:

```
mod_jobage <- lm(wage ~ jobclass + health_ins, Wage)
```

```
Call:
lm(formula = wage ~ jobclass + health_ins, data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-94.263 -24.797  -6.858  15.915 208.798

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    112.891      1.154   97.802  <2e-16 ***
jobclass2. Information  13.738      1.445    9.504  <2e-16 ***
health_ins2. No    -25.698      1.568   -16.389  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 39.12 on 2997 degrees of freedom
Multiple R-squared:  0.1215,    Adjusted R-squared:  0.1209
F-statistic: 207.3 on 2 and 2997 DF,  p-value: < 2.2e-16
```

How would you interpret the coefficients?

Model 2 – Wage, jobclass, and health_ins

- Can write this model as:

```
mod_jobage <- lm(wage ~ jobclass + health_ins, Wage)
```

```
Call:
lm(formula = wage ~ jobclass + health_ins,
    data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-94.263 -24.797  -6.858  15.915  208.798

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   112.891     1.154   97.802  <2e-16 ***
jobclass2. Information  13.738     1.445   9.504  <2e-16 ***
health_ins2. No    -25.698     1.568  -16.389  <2e-16 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 39.12 on 2997 degrees of freedom
Multiple R-squared:  0.1215,    Adjusted R-squared:  0.1209
F-statistic: 207.3 on 2 and 2997 DF,  p-value: < 2.2e-16
```

Expected mean raw wage
for industrial workers with
health insurance (i.e., when
both predictors equal their
reference group)

Model 2 – Wage, jobclass, and health_ins

- Can write this model as:

```
mod_jobage <- lm(wage ~ jobclass + health_ins, Wage)
```

```
Call:
lm(formula = wage ~ jobclass + health_ins,
    data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-94.263 -24.797  -6.858  15.915  208.798

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    112.891     1.154   97.802  <2e-16 ***
jobclass2. Information  13.738     1.445    9.504  <2e-16 ***
health_ins2. No    -25.698     1.568   -16.389  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 39.12 on 2997 degrees of freedom
Multiple R-squared:  0.1215,    Adjusted R-squared:  0.1209
F-statistic: 207.3 on 2 and 2997 DF,  p-value: < 2.2e-16
```

Difference in expected mean raw wage for information vs. industrial workers while holding health_ins constant.

Model 2 – Wage, jobclass, and health_ins

- Can write this model as:

```
mod_jobage <- lm(wage ~ jobclass + health_ins, Wage)
```

```
Call:
lm(formula = wage ~ jobclass + health_ins,
    data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-94.263 -24.797  -6.858  15.915  208.798

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    112.891      1.154   97.802  <2e-16 ***
jobclass2. Information    13.738      1.445    9.504  <2e-16 ***
health_ins2. No    -25.698      1.568   -16.389  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 39.12 on 2997 degrees of freedom
Multiple R-squared:  0.1215,    Adjusted R-squared:  0.1209
F-statistic: 207.3 on 2 and 2997 DF,  p-value: < 2.2e-16
```

Difference in expected mean raw wage for those with no health insurance vs. those with health insurance while holding jobclass constant.

Model 2 – Wage, jobclass, and health_ins

- Can write this model as:

```
mod_jobage <- lm(wage ~ jobclass + health_ins, Wage)
```

```
Call:
lm(formula = wage ~ jobclass + health_ins, data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-94.263 -24.797  -6.858  15.915  208.798

Coefficients:
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jobclass2. Information  13.738     1.445    9.504  <2e-16 ***
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Multiple R-squared:  0.1215,    Adjusted R-squared:  0.1209
F-statistic: 207.3 on 2 and 2997 DF,  p-value: < 2.2e-16
```

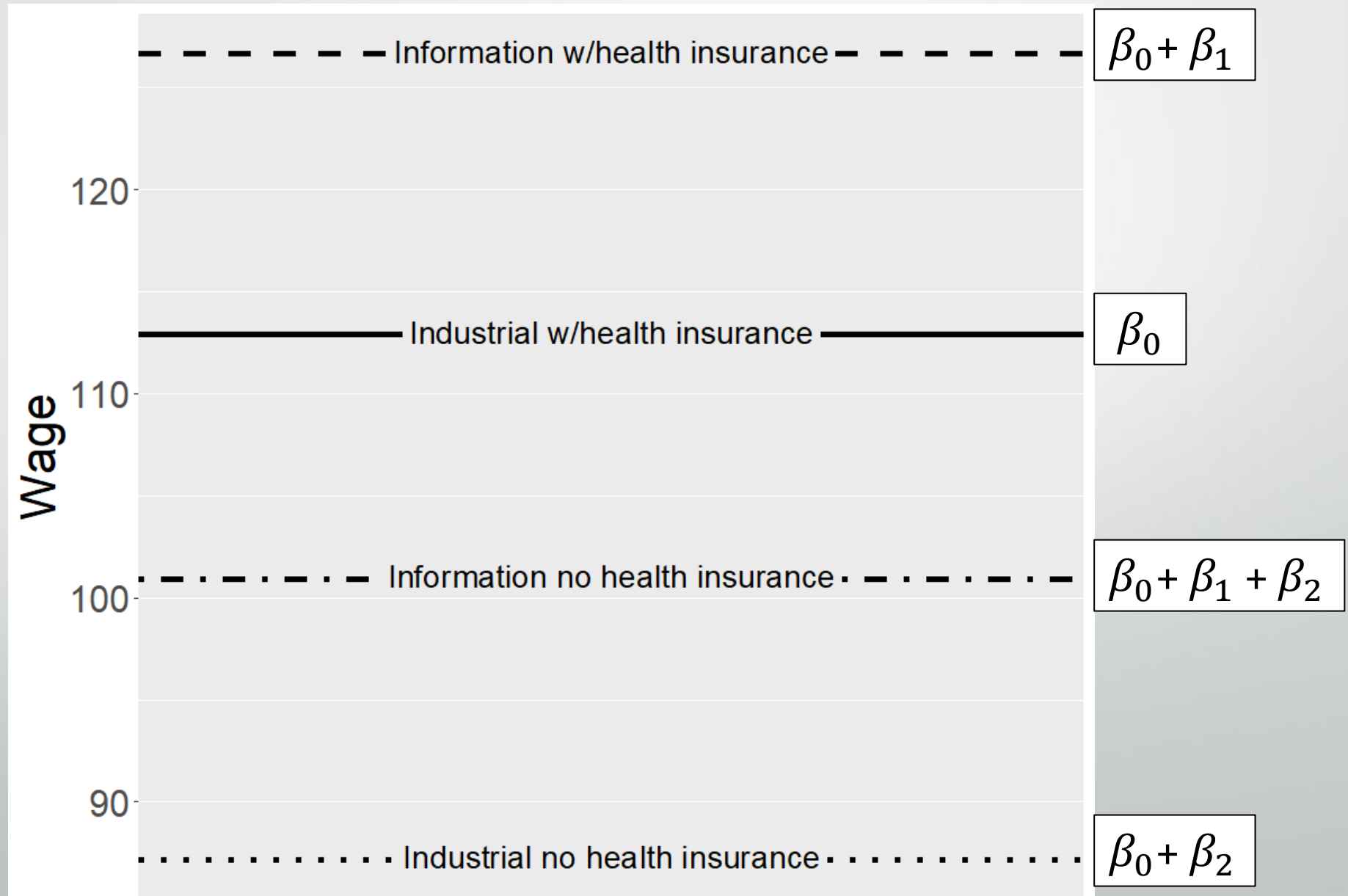
β_0

β_1

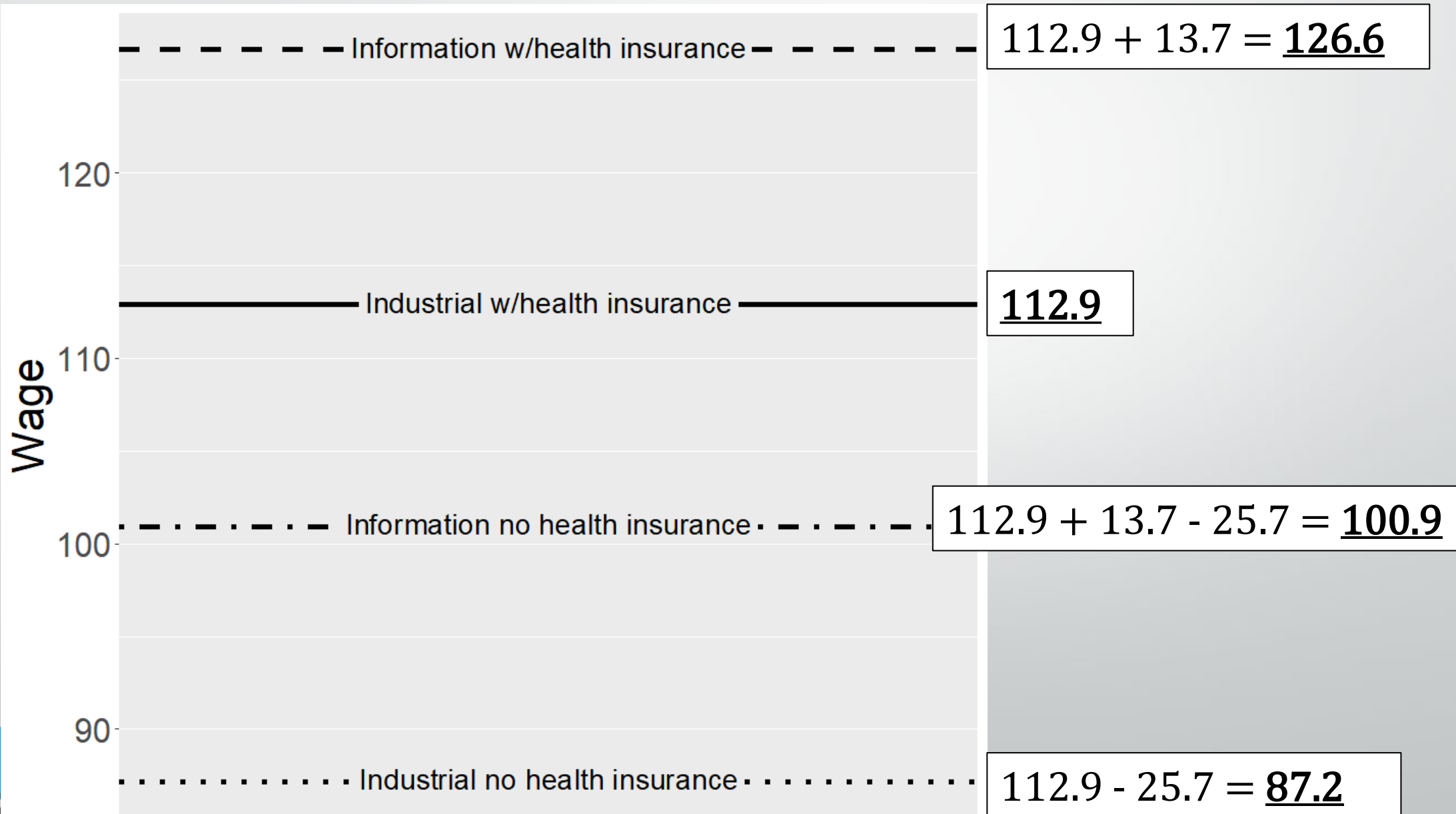
β_2

What does this
look like
graphically?

Model 2 – Wage, jobclass, and health_ins



Model 2 – Wage, jobclass, and health_ins





Predictors with More than 2 Categories

Categorical Predictors (> 2 Categories)

For a categorical predictor with f levels, consider $f-1$ dummy variables, $d_{2i}, \dots, d_{fi}$ as

- $d_{ji} = \begin{cases} \text{if } i^{th} \text{ unit is in level } j \\ \text{if } i^{th} \text{ unit is not in level } j \end{cases}$
- Level one is the reference level (i.e., other levels are compared to it)
- $y_i = \gamma_0 + \gamma_2 d_{2i} + \dots + \gamma_f d_{fi} + \epsilon_i$
- What do γ_0, γ_2 , etc. mean?

Model 3 – Wage, marital status, and age

- Can write this model as:

```
mod_maritlage <- lm(wage ~ maritl + age, Wage)
```

```
> summary(mod_maritlage)

Call:
lm(formula = wage ~ maritl + age, data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-100.97  -24.41   -5.56   15.65   219.25

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   78.65466    2.79883   28.103  < 2e-16 ***
maritl2. Married  20.81989    2.00197   10.400  < 2e-16 ***
maritl3. Widowed  -1.06328    9.40852   -0.113    0.910
maritl4. Divorced   3.93218    3.38700    1.161    0.246
maritl5. Separated  3.62631    5.67963    0.638    0.523
age             0.43212    0.07105    6.082 1.34e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.04 on 2994 degrees of freedom
Multiple R-squared:  0.0809,    Adjusted R-squared:  0.07936
F-statistic: 52.7 on 5 and 2994 DF, p-value: < 2.2e-16
```

Model 3 – Wage, marital status, and age

- Can write this model as:

```
mod_maritlage <- lm(wage ~ maritl + age, Wage)
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> summary(mod_maritlage)

Call:
lm(formula = wage ~ maritl + age, data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-100.97  -24.41   -5.56   15.65   219.25

Coefficients:
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```

How would you interpret the coefficients?

Model 3 – Wage, marital status, and age

- Can write this model as:

```
mod_maritlage <- lm(wage ~ maritl + age, Wage)
```

```
> summary(mod_maritlage)

Call:
lm(formula = wage ~ maritl + age, data = Wage)

Residuals:
    Min       1Q   Median       3Q      Max
-100.97  -24.41   -5.56   15.65   219.25

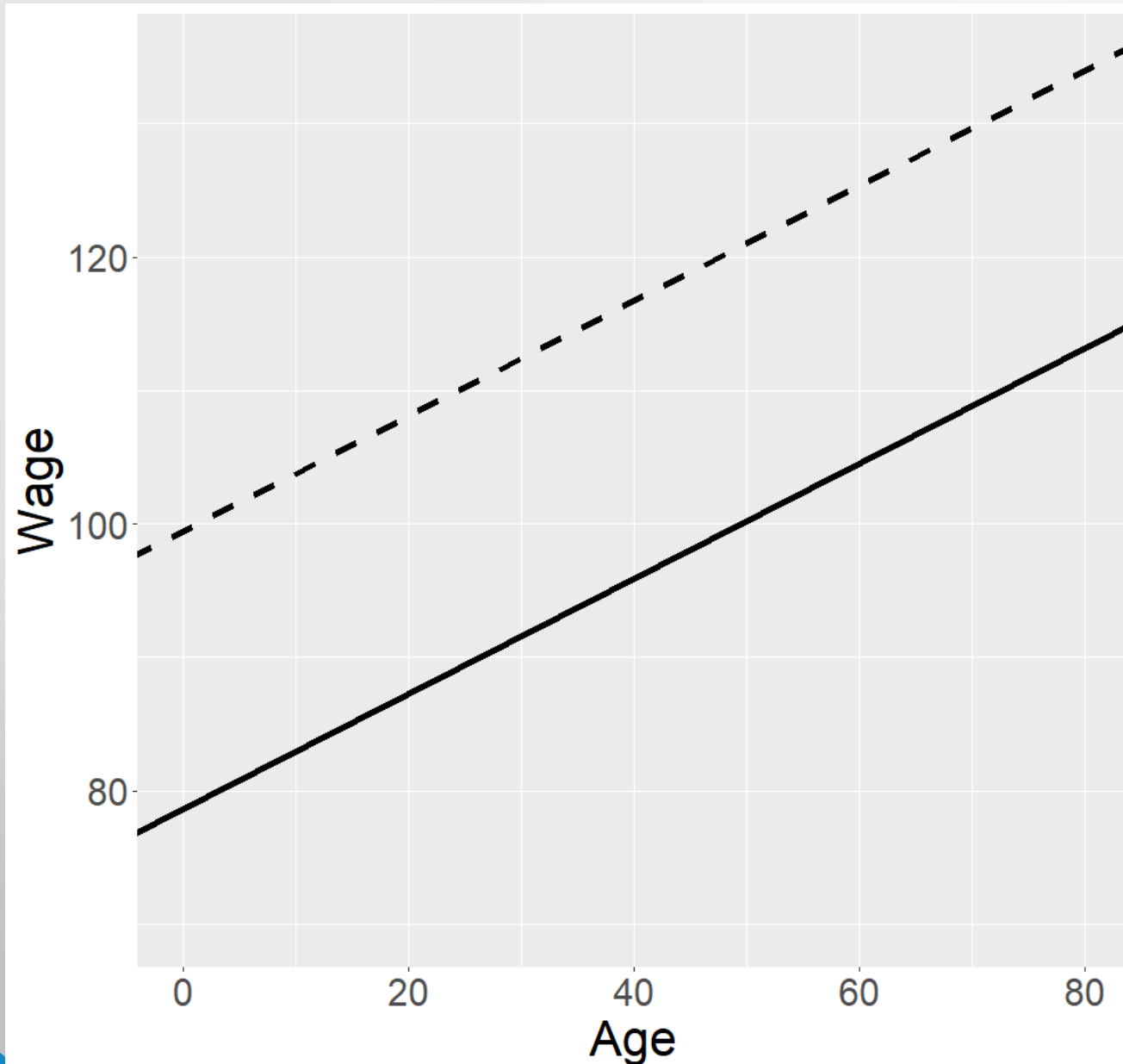
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  78.65466    2.79883   28.103  < 2e-16 ***
maritl2. Married  20.81989    2.00197   10.400  < 2e-16 ***
maritl3. Widowed  -1.06328    9.40852   -0.113    0.910
maritl4. Divorced   3.93218    3.38700    1.161    0.246
maritl5. Separated  3.62631    5.67963    0.638    0.523
age             0.43212    0.07105    6.082 1.34e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.04 on 2994 degrees of freedom
Multiple R-squared:  0.0809,    Adjusted R-squared:  0.07936
F-statistic: 52.7 on 5 and 2994 DF,  p-value: < 2.2e-16
```

How would you interpret the coefficients?

What does this look like graphically?

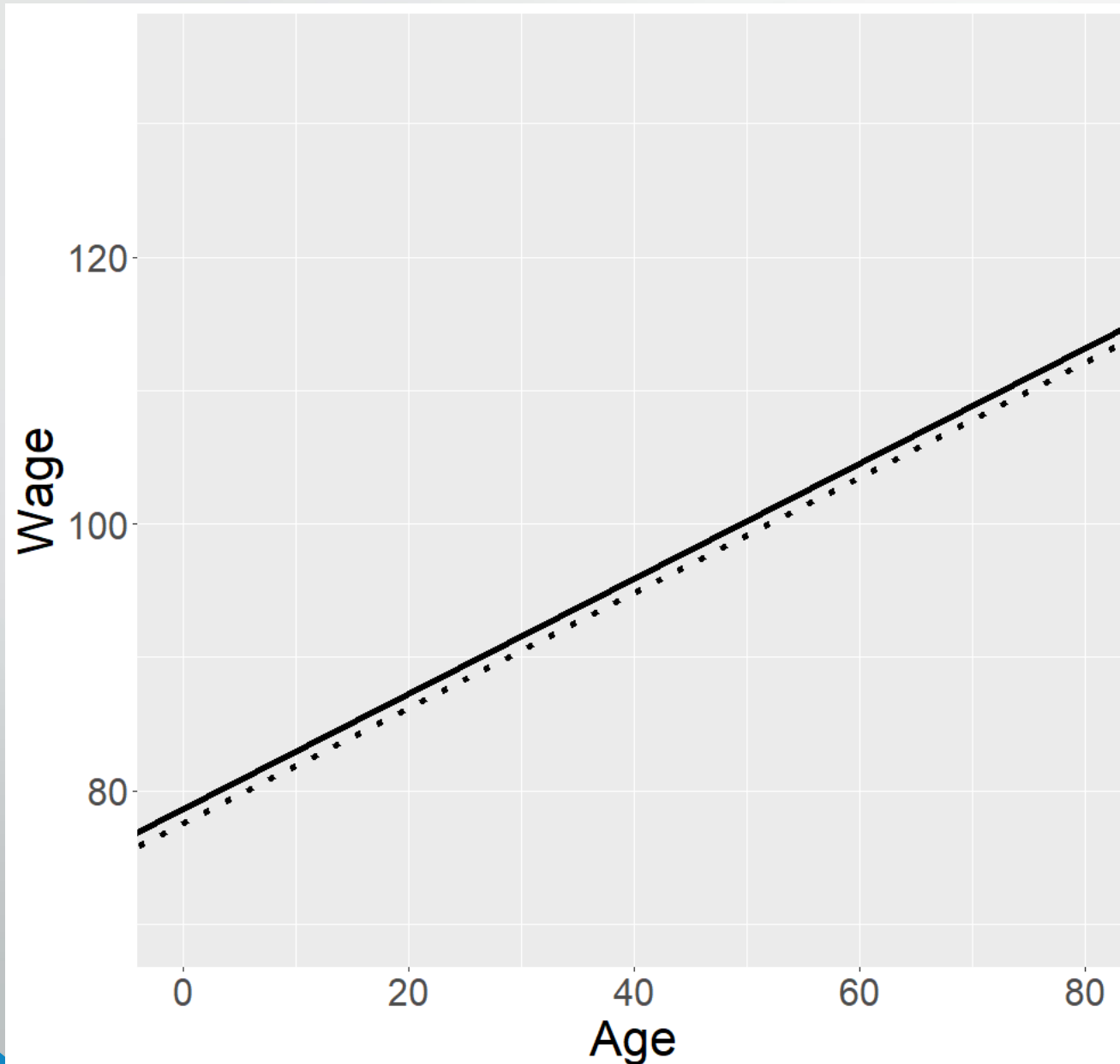
Model 3 – Wage, marital status, and age



Mean Wage for
married
participants by
age

Mean Wage for
those never
married
participants by age

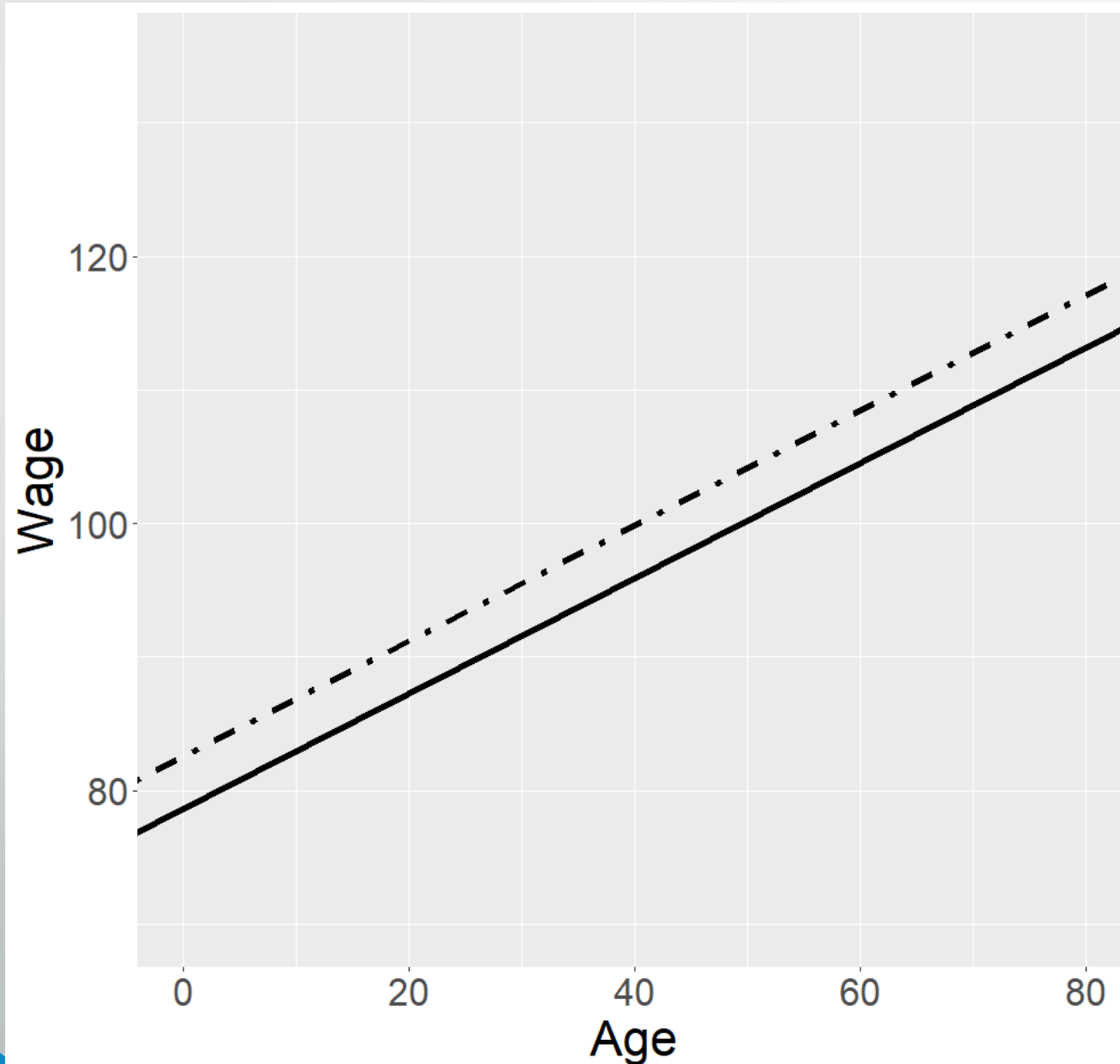
Model 3 – Wage, marital status, and age



Mean Wage for
those never
married
participants by age

Mean Wage for
widowed
participants by age

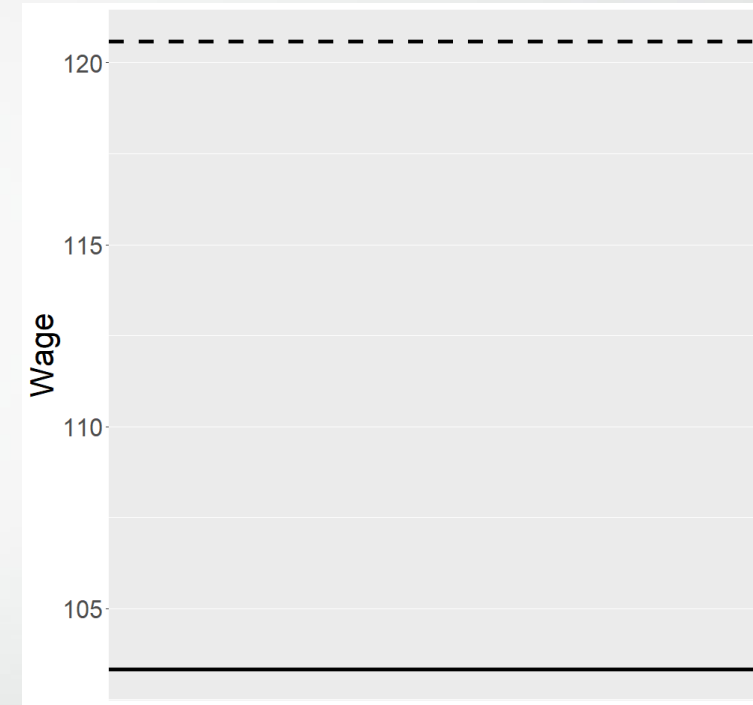
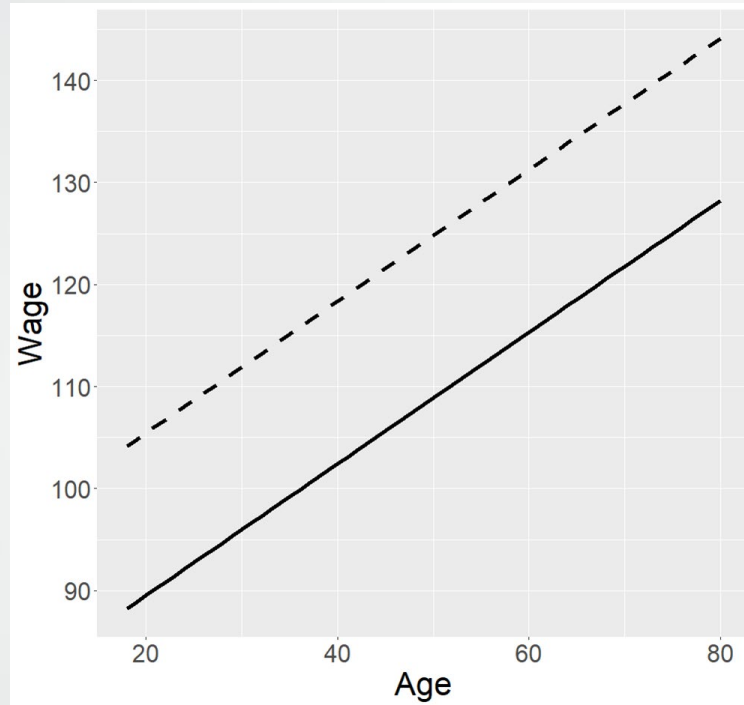
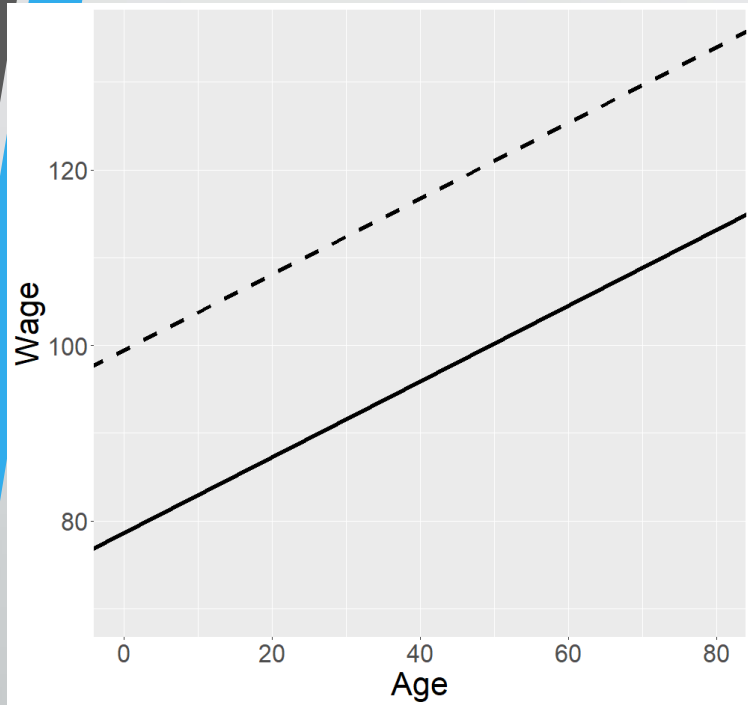
Model 3 – Wage, marital status, and age



Mean Wage for divorced participants by age

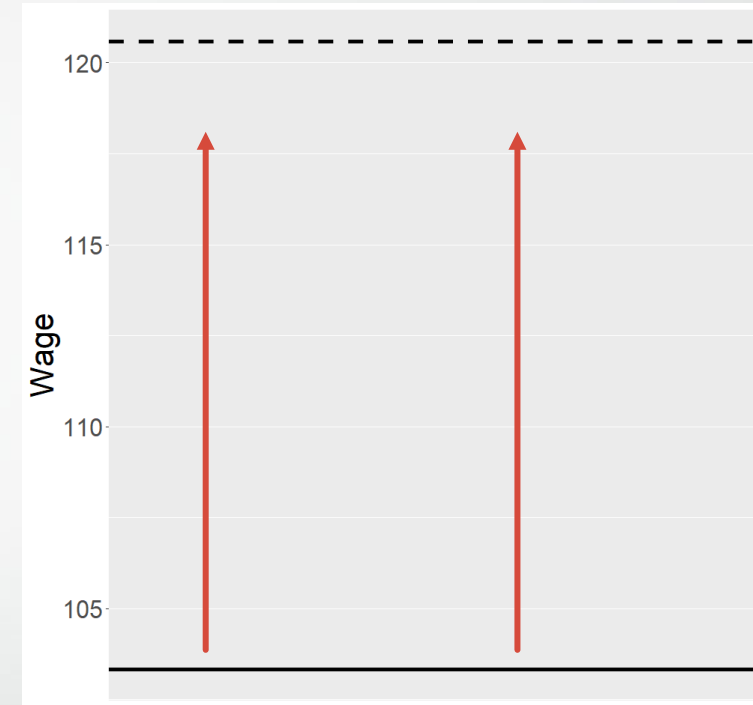
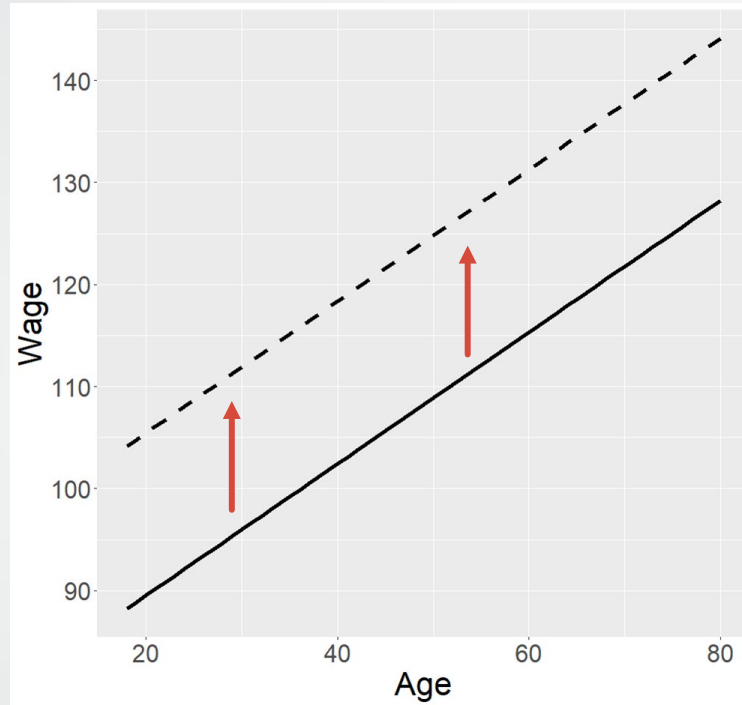
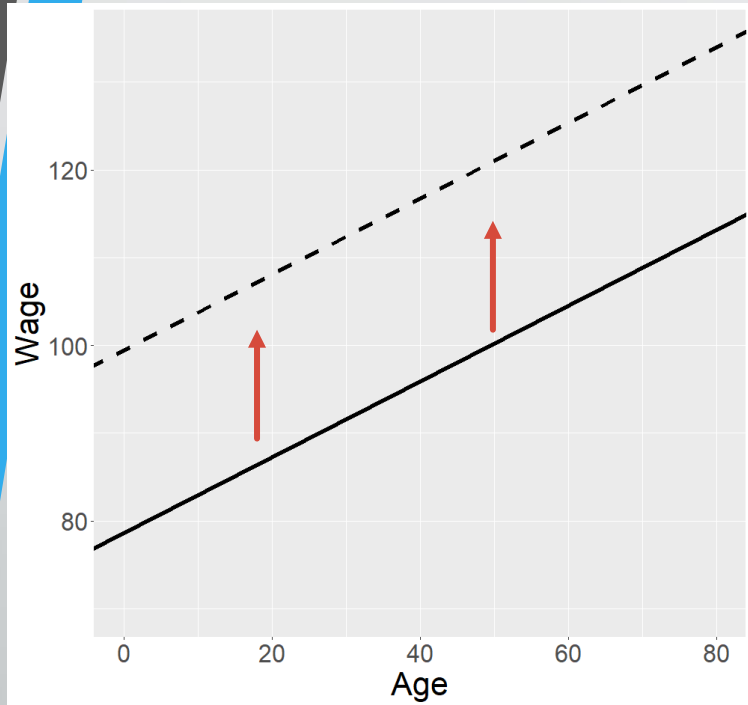
Mean Wage for those never married participants by age

Models with categorical predictors



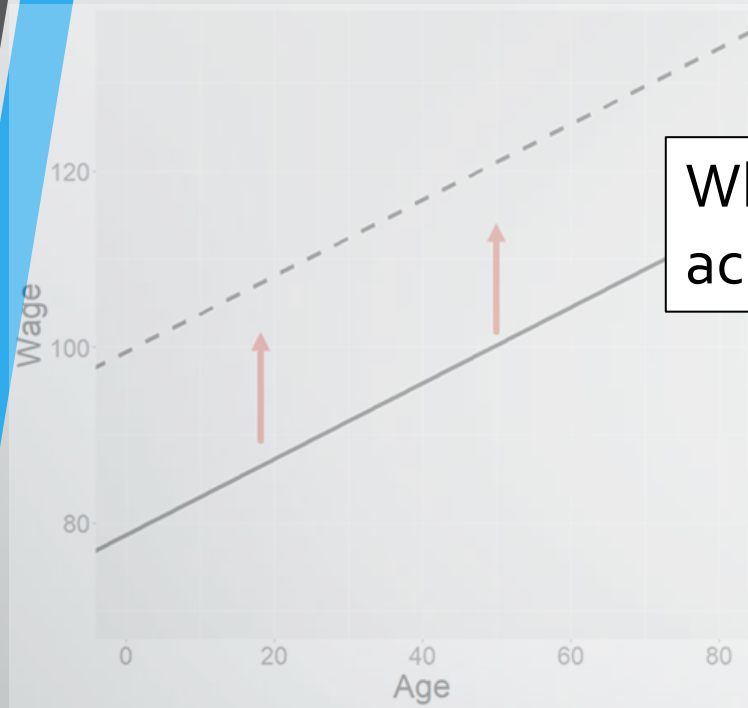
Notice anything about the regression lines from the models we've fit so far?

Models with categorical predictors

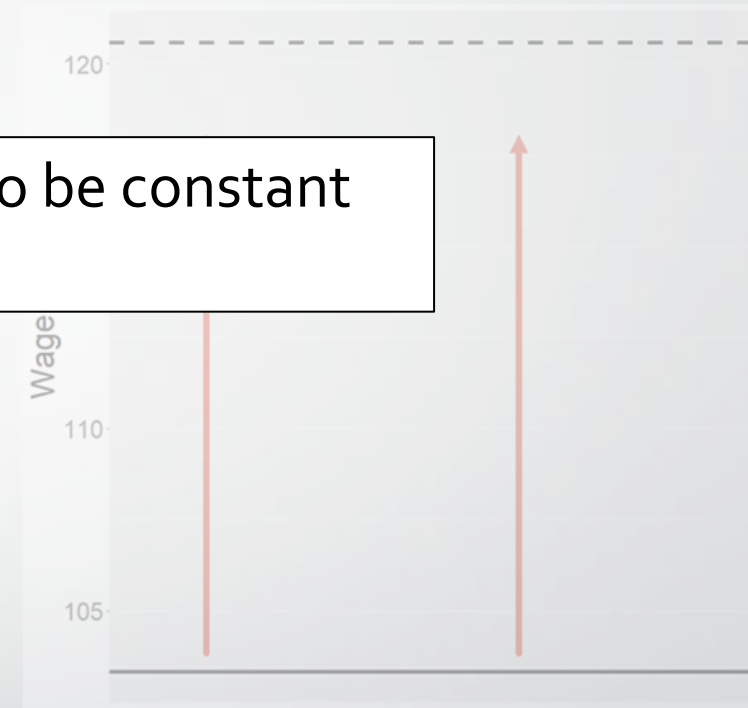
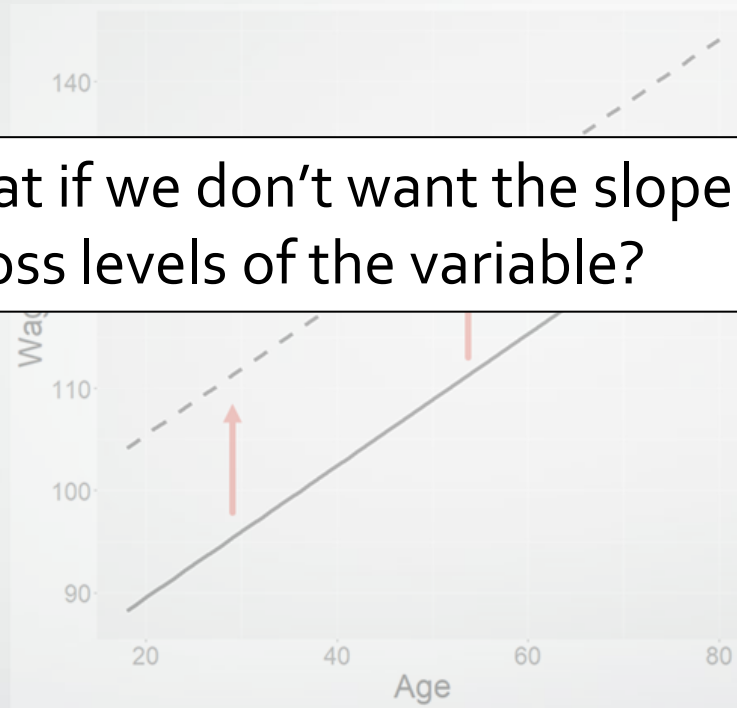


The intercept shifts (up or down) by the value of the coefficient (β or γ), but the slope stays constant.

Models with categorical predictors



What if we don't want the slope to be constant across levels of the variable?



The intercept shifts up by the value of the coefficient (β or γ), but the slope stays constant.

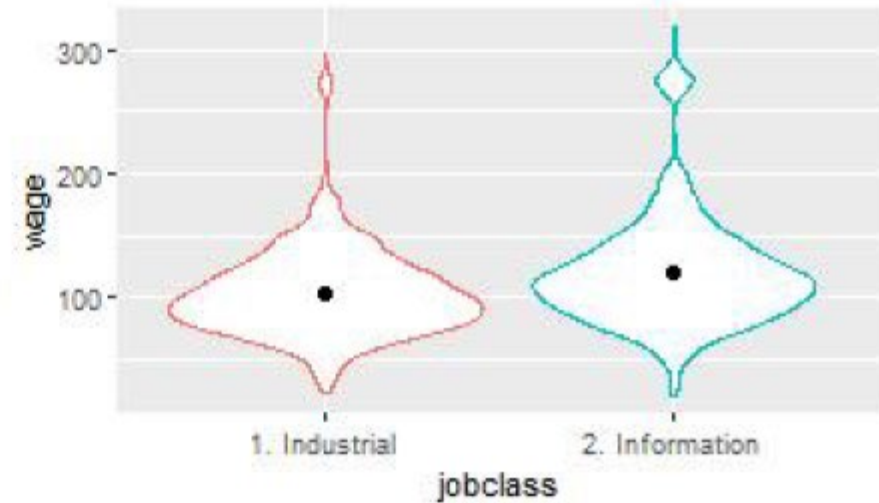
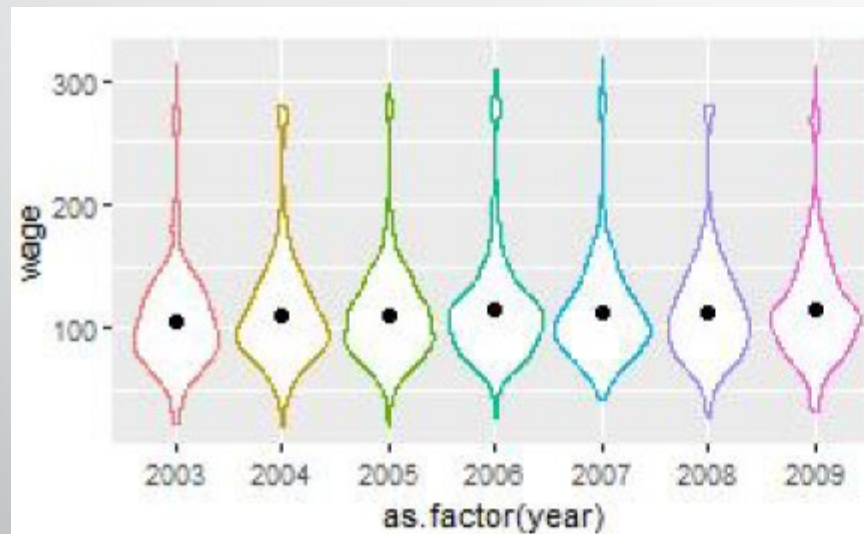


Interactions of Predictor Variables

Interactions of predictor variables

- Say we want to model `wage` as a function of `year` and `jobclass`, but don't want to assume that the slopes are the same?
- Can write this model as:

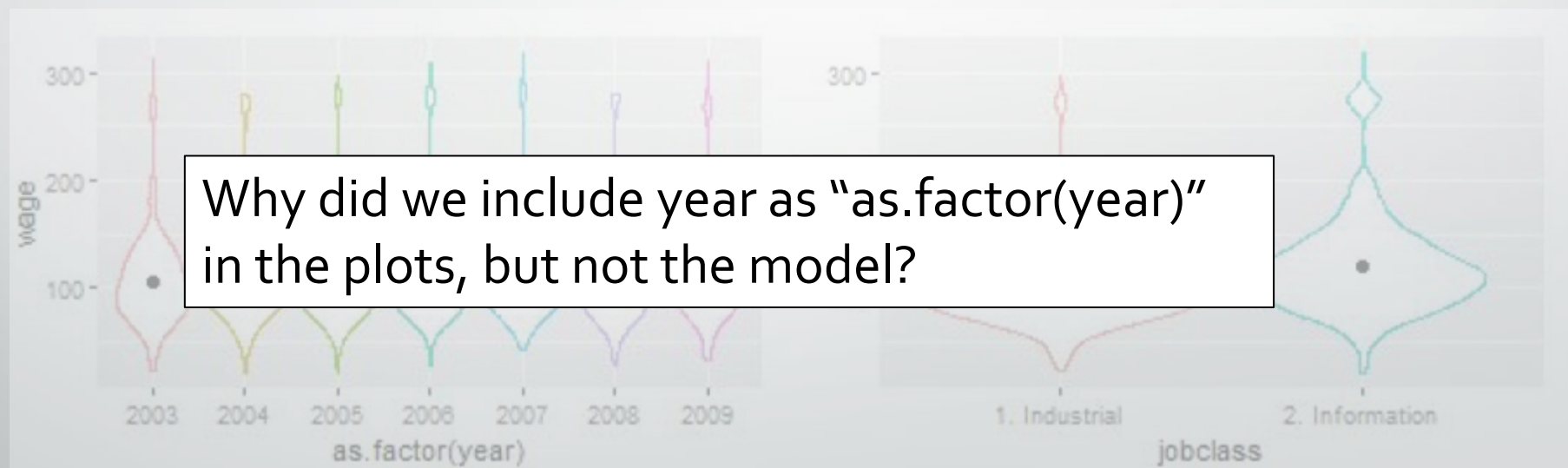
```
mod_jobyrvar <- lm(wage ~ year*jobclass, Wage)
```



Interactions of predictor variables

- Say we want to model `wage` as a function of `year` and `jobclass`, but don't want to assume that the slopes are the same?
- Can write this model as:

```
mod_jobyrvar <- lm(wage ~ year*jobclass, Wage)
```



Interactions of predictor variables

- R often makes assumptions about variables in your datasets which may or may not be accurate.
- Can check how R is classifying your variable by running `class(Wage$year)`

```
> class(Wage$year)
[1] "integer"
> |
```

We will see in lab how changing the class of a variable can impact your model.

What is an interaction? - 1

